CSE 219 — Signals and Linear Systems

Lec 5 · Sampling, Aliasing and Reconstruction

Lesson notes · Exam approach · TF practice with solutions · Quick notes (Ashraf Sir's portion)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

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Exam Map — What the TF Papers Ask — sampling questions and their weight

Lesson 9 — Impulse-Train Sampling and the Spectrum of the Sampled Signal — $X_p = (1/T)\Sigma X(j(\omega - k\omega_s))$, $P(j\omega)$, duality, FT of impulse trains

Lesson 10 — The Sampling Theorem and Aliasing — $\omega_s > 2\omega_M$, proof, oversampling vs aliasing, natural sampling

Lesson 11 — Ideal Reconstruction, ZOH and Linear Interpolation — sinc interpolation, h0, h1, comparison, real-time DAC

Lesson 12 — Practical Issues; Connection to the DFT — time-limited ⇒ not band-limited, anti-alias filter, periodic signals

Quick Notes — Sampling — merged revision notes + key diagrams

> **NOTE** Slide numbers are PDF page numbers of Lecture 5 - Sampling.pdf (47 pages, “Lec 5”); the printed slide number is one less. Natural sampling and discrete-time down-sampling (CSE 219 2022-23 Q4(b)) are textbook background (marked).

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note
2023-24 (CSE 219)	Q5(c) Compare ideal (sinc), ZOH, linear interpolation; choose for a real-time DAC (10) · Q7(c) Nyquist–Shannon theorem; $X_p(j\omega)$ satisfying/violating; why aliasing is irrecoverable (10)
2022-23 (CSE 219)	Q1(c) Band-limited x(t), $\omega_s > 2\omega_M$: show x(t) is uniquely determined by x(nT) (15) · Q4(b) $X(e^{j\omega}) = 0$ for $2\pi/9 \le \lvert\omega\rvert \le \pi$: lowest rate at which x[n] can be sampled (decimated) (10)
2021-22 (CSE 311)	Q3(b) Nyquist-rate samples all zero except g(0) = A: find g(t) (15)
2020-21 (CSE 311)	Q1(b) Can past samples foretell the future of a band-limited signal? (10) · Q3(a) Why practical signals can't be fully recovered; aliasing; how to minimise (24)
2019-20 (EEE 211)	Q7(b) $x = \sin(\omega_B t/2)/(\pi t) \times \Sigma\delta(t - n\pi/\omega_B) \rightarrow h = \sin(3\omega_B t/2)/(\pi t)$: y(t) (20) · Q8(a) FT of an impulse train is an impulse train (15)
2019-20 (CSE 311)	Q3 FT of $2B\,\mathrm{sinc}(2\pi Bt)$ / $B\,\mathrm{sinc}^2(\pi Bt)$; spectrum when sampled at 1.5× Nyquist; general sampled spectrum (35)
2018-19 (EEE 211)	Q3(a) FT of an impulse train is an impulse train; sketch both (10) · Q3(b) FT of $\sin 1000\pi t + \cos 1200\pi t$; sampling period for no aliasing (10)
2018-19 (CSE 311)	Q1(a) g(t) band-limited to 50 Hz, samples 0.01 s apart all 0 except g(0)=1: g(t), G(f); FT of δ (21) · Q3(b) Cause of aliasing, its problems, how compensated (20)
2017-18 (EEE 211)	Q2(c) Natural sampling with a rectangular pulse train: spectrum; recoverable? (15)
2016-17 (EEE 211)	Q6(c) x·p (impulse train $n\pi/\omega_B$) → h = $\sin(3\omega_B t/2)/(\pi t)$: spectra p, z, h, y (12)
2016-17 (CSE 311)	Q2(b) State & prove the sampling theorem; anti-alias filter; why sample above Nyquist (16)
2015-16 (EEE 301)	Q6(b) z = x·y (150 Hz, 350 Hz): max T_s; spectra for T_s = 0.5 ms, 2 ms (12)
2014-15 (EEE 301)	Q6(b) Natural sampling with rectangular pulses: spectrum; recoverable? (15)
2014-15 (CSE 311)	Q8(a) FT of an impulse-sampled signal; sampling theorem (15) · Q2(c) [L-3] Aliasing effect; how eliminated (10)
2013-14 (EEE 301)	Q8(a) Impulse sampling (T = 1/8 ms) of a 5 kHz signal, filter: P, X_s, Y; recoverable? (17)
2013-14 (CSE 311)	Q2(a) State and prove Nyquist's sampling theorem (13) · Q2(b) Practical issues in sampling and reconstruction; how handled (10)
2012-13 (EEE 301)	Q6(b) Sampler → h → squarer: X, P, Z, H, Y, S (15)
2012-13 (CSE 311)	Q8(a) Prove the minimum sampling frequency is 2B (15)
2011-12 (EEE 301)	Q2(b) Alternating-sign impulse train sampling: recovery system; X_p, Y (18)
2010-11 (EEE 301)	Q2(b) x = $\sin(\omega_B t/2)/(\pi t)$ × impulse train: X, P, X_s, h, H, Y (15)
2010-11 (CSE 311)	Q6(c) Samples at 2B through ideal LPF (bandwidth B) give an output ∝ f(t) (12)

EXAM FOCUS Sampling is asked in every paper: 'state and prove the sampling theorem' (CSE 311 2012-13, 2013-14, 2016-17; CSE 219 2022-23 Q1(c) 15 marks, 2023-24 Q7(c) 10), spectrum sketches of an impulse-sampled signal (EEE every year) and aliasing explanations. Know the one-line derivation $X_p = (1/T)\Sigma X(j(\omega - k\omega_s))$ and the sketch of non-overlapping vs overlapping copies.

Lesson 9 — Impulse-Train Sampling and the Spectrum of the Sampled Signal

Lec 5 sl. 4–17

Many signals give the same samples (sl. 5), so recovery needs a condition. Model sampling as multiplication by an impulse train (sl. 6):

$$x_p(t) = x(t)p(t) = \Sigma\, x(nT)\delta(t - nT), \quad p(t) = \Sigma\, \delta(t - nT), \quad \omega_s = 2\pi/T$$

Two roads to X_p(jω) (sl. 8–17)

- Road 1 — transform the product: X_p(jω) = Σ x(nT)$e^{-j\omega nT}$: correct (it is the DTFT after ωT → ω) but X(jω) does not appear — uninterpretable.
- Road 2 — multiplication in time = convolution in frequency: X_p = (1/2π)X(jω) * P(jω). Needs P(jω).

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Duality:  x(t) ↔ X(jω) ⇒ X(jt) ↔ 2πx(-ω)       (from x(t) = (1/2π)∫X e^{jωt}dω: 2πx(-t) =
∫X(jω)e^{-jωt}dω, rename)
Multiplication ↔ convolution:  u(t)v(t) ↔ (1/2π)U * V   (duality of the convolution property)
P(jω): p is T-periodic ⇒ Fourier series p = (1/T)Σ e^{jkω_s t} (c_k = (1/T)∫δ(t)e^{-jkω_s t}dt
= 1/T);
       e^{jkω_s t} ↔ 2πδ(ω - kω_s)  ⇒  P(jω) = (2π/T)Σ δ(ω - kω_s)   (an impulse train is an
impulse train)
Convolving with δ(ω - kω_s) just shifts: X * δ(ω - kω_s) = X(j(ω - kω_s))
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$$X_p(j\omega) = (1/2\pi)\, X(j\omega) * (2\pi/T)\Sigma\delta(\omega - k\omega_s) = (1/T)\, \Sigma_{k=-\infty}^{\infty}\, X(j(\omega - k\omega_s))$$

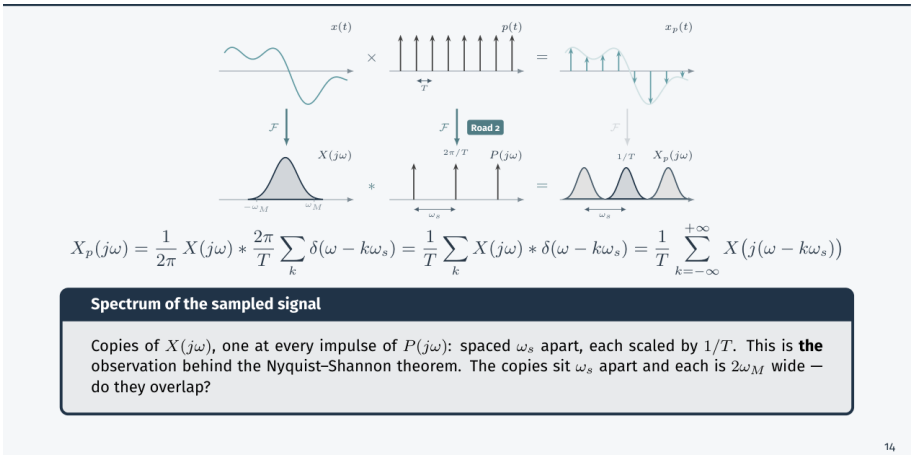


Figure 9.1 — Sampling replicates X(jω) at every kω_s, scaled by 1/T (Lec 5, sl. 17)

FT of an Impulse Train (also asked directly)

Σ δ(t − nT) has Fourier series coefficients 1/T for every k, so its FT is (2π/T)Σδ(ω − 2πk/T): impulses of area 2π/T at multiples of ω_s. Closer samples (small T) give sparser, higher spectral lines.

MEMORIZE Spectrum of the sampled signal = the original spectrum repeated every ω_s and divided by T. Copies are 2ω_M wide and ω_s apart.

Exam Approach

1. Sketch X(jω) with its edge ω_M.
2. Write P(jω) = (2π/T)Σδ(ω − kω_s) and the factor 1/(2π).

3. Draw copies at kω_s with height scaled by 1/T.
4. Decide overlap: ω_s − ω_M vs ω_M.

Practice — Lesson 9

(a) Actual — CSE 311 2019-20 Q3. (b) Actual — EEE 211 2018-19 Q3(a) and 2019-20 Q8(a). (c) Actual — EEE 211 2019-20 Q7(b), 2016-17 Q6(c). (d) Actual — EEE 301 2013-14 Q8(a), 2011-12 Q2(b), 2010-11 Q2(b), 2012-13 Q6(b). (e) Actual — CSE 311 2014-15 Q8(a).

(a) FT of g(t) = 2B sinc(2πBt) [sinc x = sin x/x]; spectrum after sampling at 1.5 × Nyquist; general expression for the sampled spectrum at rate f_s. (35)

(b) Show that the FT of an impulse train is an impulse train; sketch both. (10–15)

(c) x(t) = sin(ω_Bt/2)/(πt), p(t) = Σδ(t − nπ/ω_B), z = xp, y = z * h with h = sin(3ω_Bt/2)/(πt). Find y(t). (20; 12)

(d) Impulse sampling of a band-limited signal followed by filters (alternating sign train, squarer, 1/8 ms period).

(e) FT of the impulse-sampled signal G_δ(f); the sampling theorem. (15)

✓ Solution

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(a) g(t) = 2B sinc(2πBt) = sin(2πBt)/(πt) ↔ G(f) = rect(f/2B) (height 1, |f| < B). Nyquist rate
2B, so f_s = 3B:
    G_s(f) = f_s Σ_k G(f − k f_s) = 3B Σ rect((f − 3Bk)/2B): rectangles of height 3B, width 2B,
every 3B (gaps of B).
    (for Bsinc²(πBt): G = triangle of height 1 and half-width B ⇒ Nyquist 2B again; at 3B:
triangles every 3B, height 3B, non-overlapping)
    general: G_δ(f) = f_s Σ_{n=-∞}^{∞} G(f − n f_s)   (from g_δ(t) = g(t)Σδ(t − nT_s), Σδ(t −
nT) = f_s Σ e^{j2πn f_s t})
(b) Σδ(t − nT) = (1/T)Σ e^{jkω_s t} ⇒ FT (2π/T)Σδ(ω − kω_s): impulses of area 2π/T at 0, ±ω_s,
±2ω_s …
(c) X = rect(|ω| < ω_B/2) (height 1); P = (2ω_B)Σδ(ω − 2ω_B k)  [T = π/ω_B ⇒ ω_s = 2ω_B]; Z =
(1/2π)X * P = ω_B/π Σ X(ω − 2kω_B)/…
    copies of height ω_B/π at multiples of 2ω_B, each of width ω_B (no overlap: ω_B < 2ω_B −
ω_B/2... gaps exist);
    h passes |ω| < 3ω_B/2 ⇒ only the k = 0 copy (the next starts at 2ω_B − ω_B/2 = 1.5ω_B)  ⇒
    Y = (ω_B/π)·X ⇒ y(t) = (ω_B/π)x(t) = ω_B sin(ω_Bt/2)/(π²t)
(d) same picture: X_s = (1/T)ΣX(ω − kω_s); with ω_s > 2ω_M copies are separated, an LPF of
cutoff ω_s/2 and gain T recovers x.
    Alternating-sign train p = Σ(−1)ⁿδ(t − nT): period 2T, P = (2π/2T)·Σ_{odd k} 2·δ(ω − kπ/T)
(odd harmonics only) ⇒ copies
    centred at ±ω_s/2, ±3ω_s/2…: recovery requires a band-pass filter at ω_s/2 followed by
demodulation by cos(ω_s t/2).
(e) g_δ(t) = Σ g(nT_s)δ(t − nT_s) = g(t)·Σδ(t − nT_s) ⇒ G_δ(f) = G(f) * f_s Σδ(f − n f_s) =
f_s ΣG(f − n f_s)
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Lesson 10 — The Sampling Theorem and Aliasing

Lec 5 sl. 19–22; natural sampling: textbook

Band-limited x: X(jω) = 0 for |ω| > ω_M. Then x(t) is uniquely determined by x(nT) if ω_s > 2ω_M (ω_s = 2π/T)
Nyquist rate = 2ω_M (f_s > 2f_M); T < π/ω_M

Proof (exam-ready)

1. X_p(jω) = (1/T)ΣX(j(ω − kω_s)) (Lesson 9): copies of X at kω_s, each [kω_s − ω_M, kω_s + ω_M].
2. Neighbouring copies stay apart iff ω_M < ω_s − ω_M ⇔ ω_s > 2ω_M.
3. Then the central copy is an untouched X(jω)/T; an ideal LPF H = T on |ω| < ω_c, ω_M < ω_c < ω_s − ω_M, recovers X exactly, hence x(t) (Lesson 11 gives x(t) = Σx(nT)sinc((t − nT)/T)).
4. If ω_s < 2ω_M the copies overlap and add; the overlapped values cannot be separated, so no filter can recover X ⇒ x(t) is not uniquely determined (two different band-limited... signals give identical samples).



Figure 10.1 — Undersampling: the copies overlap and high frequencies fold into the baseband (Lec 5, sl. 22)

Case	Condition	Result
Oversampling	ω_s > 2ω_M	copies separated by a guard band; recoverable with an LPF
Critical sampling	ω_s = 2ω_M	copies touch; works only in the limit (sinc kernel), impractical
Undersampling	ω_s < 2ω_M	overlap = **aliasing**: frequency ω appears as ω − kω_s; irreversible

Aliasing and Practical Cures

- Cause: sampling too slowly (or a signal not band-limited): spectral copies overlap, high frequencies masquerade as low ones (e.g. wagon wheels, a 1.8 kHz tone sampled at 2.8 kHz appears at 1 kHz).
- Problems: distortion that cannot be undone; the original signal is lost.
- Cures: an analog **anti-aliasing low-pass filter** before sampling (cutoff ≤ f_s/2); sample above Nyquist (oversample, which also eases reconstruction filter design); a guard band.
- Why practical signals always alias a little: real signals are time-limited, so their spectra are not exactly band-limited (Lesson 12).

Discrete-Time Sampling (textbook)

For a sequence, $X(e^{j\omega}) = 0$ for $2\pi/9 \le |\omega| \le \pi$, i.e. bandlimited to $W = 2\pi/9$. Keeping every M-th sample (rate 1/M of the original) is alias-free iff $MW \le \pi \Rightarrow M \le 4.5 \Rightarrow M = 4$ (largest integer): the lowest rate is the original rate/4 (a sample every 4 steps): bandwidth $2\pi/9 \rightarrow$ spread to $8\pi/9 < \pi$.

Natural Sampling (textbook)

Multiply by a train of rectangular pulses of width τ, period T: p(t) has Fourier coefficients $c_k = (\tau/T)\text{sinc}(k\tau/T)$, so $X_s(\omega) = \Sigma\, c_k\, X(\omega - k\omega_s)$: the copies have **different heights** (sinc-weighted) but the same shape — x is recoverable with an LPF if $\omega_s > 2\omega_M$, only scaled by $c_0 = \tau/T$.

Exam Approach

1. Sketch $X(j\omega)$; mark ω_M; find ω_s and compare with $2\omega_M$.
2. Draw copies at $\pm\omega_s$, $\pm2\omega_s$ with height 1/T.
3. Aliasing: show the overlap region; cure: anti-aliasing filter / higher f_s.
4. Multiple signals: use the highest frequency present (bandwidth of a product = sum of the bandwidths).

Practice — Lesson 10

(a) Actual — CSE 219 2022-23 Q1(c). (b) Actual — CSE 219 2023-24 Q7(c). (c) Actual — CSE 311 2012-13 Q8(a), 2013-14 Q2(a), 2016-17 Q2(b). (d) Actual — CSE 311 2018-19 Q3(b), 2014-15 Q2(c) [L-3]. (e) Actual — CSE 219 2022-23 Q4(b). (f) Actual — EEE 211 2018-19 Q3(b), EEE 301 2015-16 Q6(b). (g) Natural sampling: EEE 211 2017-18 Q2(c), EEE 301 2014-15 Q6(b).

(a) Band-limited x(t) with $\omega_s > 2\omega_M$; analyze how x(t) is uniquely determined by its samples x(nT). (15)

(b) State the Nyquist–Shannon theorem; analyze $X_p(j\omega)$ when ω_s satisfies/violates it; sketch; why is perfect reconstruction impossible after aliasing? (10)

(c) State and prove the sampling theorem; why in practice is the signal band-limited first and why is the rate above Nyquist? Prove the minimum sampling frequency is 2B. (13–16)

(d) Cause of aliasing, its problems and compensation. (20)

(e) A sequence has $X(e^{j\omega}) = 0$ for $2\pi/9 \le |\omega| \le \pi$. Lowest rate to sample it for perfect reconstruction? (10)

(f) $x = \sin 1000\pi t + \cos 1200\pi t$: sampling period for no aliasing. $z = xy$ with x, y band-limited to 150 Hz and 350 Hz: maximum T_s; spectra for $T_s = 0.5$ ms and 2 ms. (10; 12)

(g) Natural sampling with a rectangular pulse train: spectrum; can x be recovered?

✓ Solution

(a) Proof steps 1–3 above; uniqueness: if two band-limited signals had equal samples, their sampled spectra $(1/T)\Sigma X_1(j(\omega - k\omega_s))$ and $(1/T)\Sigma X_2(\ldots)$ would be equal, and with no overlap the central copies $X_1 = X_2 \Rightarrow x_1 = x_2$.

(b) Theorem (steps 1–4). Satisfied: copies spaced ω_s apart, guard band $\omega_s - 2\omega_M$, LPF recovers x. Violated: overlap, aliasing; the overlapped region is a sum of two or more frequency components, an information loss that no linear filter can invert (in the overlap $X_p(j\omega)$ cannot tell which copy each part came from).

(c) Proof as above (CSE 311 papers: $X_p(f) = f_s\Sigma G(f - nf_s)$; LPF of bandwidth B recovers G if $f_s \ge 2B$ — the boundary $f_s = 2B$ is the minimum). (i) A band-limiting filter before sampling removes energy above $f_s/2$ that would alias. (ii) A rate above Nyquist leaves a guard band so a realizable (non-ideal, finite-slope) reconstruction filter can separate the copies.

(d) Cause: copies of the spectrum overlap because $f_s < 2f_M$ (or the signal has components above $f_s/2$). Effects: a component at f appears at $|f - kf_s|$ in the baseband (e.g. 1.8 kHz at f_s = 2.8 kHz → 1 kHz), irreversibly distorting the signal. Compensation: anti-aliasing LPF before the sampler, higher sampling rate, guard band; after sampling nothing can be undone.

(e) $W = 2\pi/9$; down-sampling by M keeps no aliasing iff $MW \le \pi \Rightarrow M \le 4.5 \Rightarrow$ **M = 4**: the sequence can be sampled (decimated) at ¼ of its rate; the lowest rate is 1 sample per 4 (new bandwidth $8\pi/9 < \pi$).

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(f) sin 1000πt: f = 500 Hz; cos 1200πt: 600 Hz ⇒ f_M = 600 Hz ⇒ f_s > 1200 Hz ⇒ T < 1/1200 s
= 0.833 ms
    z = xy: bandwidth 150 + 350 = 500 Hz ⇒ T_s,max = 1/(2·500) = 1 ms
    T_s = 0.5 ms (f_s = 2 kHz): copies every 2 kHz, each ±500 Hz wide ⇒ no overlap (guard 1
kHz)
    T_s = 2 ms (f_s = 500 Hz): copies every 500 Hz overlap (width 1000 Hz) ⇒ aliasing, flat
sum between copies
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(g) Pulse train coefficients $c_k = (\tau/T)\mathrm{sinc}(k\tau/T)$; $X_s(\omega) = \Sigma X\, c_k\, X(\omega - k\omega_s)$: copies of the same shape with heights $c_k \cdot X$; recoverable by an LPF of gain T/τ if $\omega_s > 2\omega_M$.

Lesson 11 — Ideal Reconstruction, ZOH and Linear Interpolation

Lec 5 sl. 24–36

1. Ideal Reconstruction (sl. 24–27)

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H(jω) = T for |ω| < ω_s/2 (= π/T), 0 otherwise  (ω_M < ω_c < ω_s − ω_M)
h(t) = (1/2π)∫_{−π/T}^{π/T} T e^{jωt}dω = sin(πt/T)/(πt/T) = sinc(t/T)   (sinc x = sin πx/πx)
x_r(t) = x_p * h = Σ x(nT) sinc((t − nT)/T)   (sum of shifted sincs)
h is non-causal (non-zero for t < 0) and decays slowly ⇒ unrealizable ⇒ approximations
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2. Zero-Order Hold (sl. 28–31)

Hold each sample for T: $h_0(t) = 1$ on [0, T] (a staircase; every real DAC). Each sample y(nT) contributes $y(nT)h_0(t - nT)$, which must be 1 from nT to (n + 1)T and 0 elsewhere.

$$\mathbf{H_0(j\omega) = \int_0^T e^{-j\omega t}dt = (1 - e^{-j\omega T})/(j\omega) = T\, e^{-j\omega T/2}\, \mathrm{sinc}(\omega T/2\pi)}$$

Magnitude $T|\mathrm{sinc}(\omega T/2\pi)|$: droops inside the passband $|\omega| < \omega_s/2$ (about −3.9 dB at $\omega_s/2$) and does not cut out the aliasing copies (non-zero beyond $\omega_s/2$). Phase $-\omega T/2$: a pure delay of T/2 — causal.

3. Linear Interpolation / First-Order Hold (sl. 32–35)

Join adjacent samples by straight lines: the weight of y(nT) must rise 0 → 1 over [(n − 1)T, nT] and fall 1 → 0 over [nT, (n + 1)T] ⇒ h_1 = triangle on [−T, T], peak 1. It equals $(1/T)\cdot[h_0(t + T/2) * h_0(t + T/2)]$.

$$\mathbf{H_1(j\omega) = (1/T)[T\,\mathrm{sinc}(\omega T/2\pi)]^2 = T\,\mathrm{sinc}^2(\omega T/2\pi)}$$

(real, no delay — the triangle is non-causal; delaying it by T makes it causal)

	Ideal (sinc)	ZOH (rect)	Linear interp. (triangle)
h(t)	sinc(t/T), ∞ both sides	1 on [0, T]	triangle on [−T, T]
H(jω)	T rect(|ω| < ω_s/2)	$T\, e^{-j\omega T/2}$ sinc(ωT/2π)	T sinc²(ωT/2π)
causal?	no	yes	no (needs one sample ahead; delay T makes it causal)
|H| at ω_s/4 (T = 1)	1	0.90	0.81
out-of-band leakage	none	large (sinc sidelobes)	small (sinc² falls faster)
use	theory, offline	real-time DACs	smoother offline/graphics

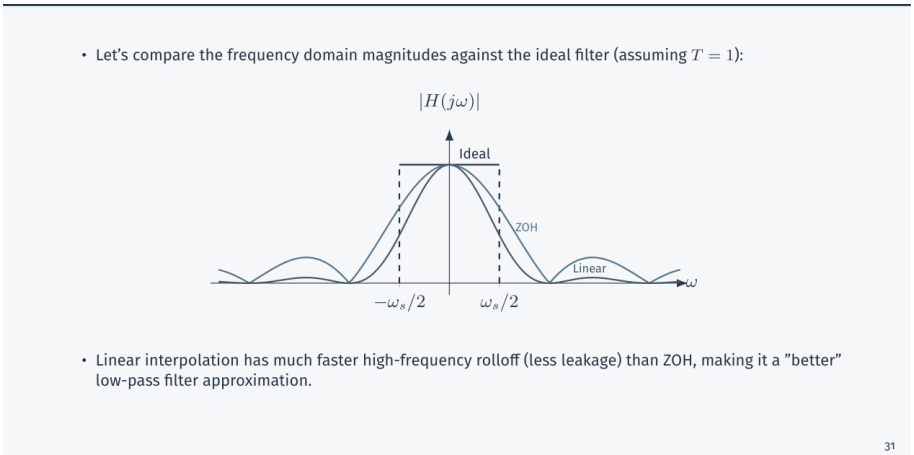


Figure 11.1 — Magnitude responses of ideal, ZOH and linear interpolation (Lec 5, sl. 36)

EXAM-READY ANSWER Real-time DAC (CSE 219 2023-24 Q5(c)(ii)): choose the **zero-order hold**. It is causal, needs only the current sample (no delay buffer), is a simple hold circuit inside the DAC, and its droop/leakage can be corrected with an analog reconstruction filter. The ideal sinc is non-causal and infinitely long; linear interpolation needs the next sample (one-sample delay buffer), so it is not directly real-time.

Exam Approach

1. Write h(t) for each method and the reason it has that shape (weight of one sample).
2. Derive H(jω) by Fourier integration (rect → sinc), triangle = rect * rect.
3. Compare causality, passband flatness and out-of-band leakage.
4. State a one-line advantage and drawback of each.

Practice — Lesson 11

(a) Actual — CSE 219 2023-24 Q5(c). (b) Actual — CSE 311 2018-19 Q1(a), 2021-22 Q3(b), 2010-11 Q6(c), 2020-21 Q1(b).

(a) An engineer must convert stored samples to an analog signal using an ideal sinc low-pass filter, a ZOH, or linear interpolation; the hardware cannot do non-causal filtering and no delay buffer is allowed. (i) Compare the three impulse responses. (ii) Which should he choose? (5 + 5)

(b) g(t) is band-limited to 50 Hz; samples at 0.01 s: g(0) = 1, all others 0. Find g(t) and G(f) and its characteristics. Same with Nyquist-rate samples zero except g(0) = A. Samples at 2B through an ideal LPF of bandwidth B give an output ∝ f(t). Can the past samples foretell the future?

✓ Solution

(a) (i) Ideal: h = sinc(t/T), two-sided and infinite ⇒ non-causal. ZOH: h_0 = rect on [0, T] ⇒ causal, finite. Linear: h_1 = triangle on [−T, T] ⇒ finite but non-causal. (ii) ZOH (see EXAM-READY ANSWER).

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(b) samples g(0.01k) = δ[k]: f_s = 100 Hz = 2 × 50 Hz (Nyquist rate), so g = Σδ[k]sinc((t −
0.01k)/0.01) = sinc(100t)
    = sin(100πt)/(100πt);  G(f) = 0.01·rect(f/100) (flat 0.01 on |f| < 50 Hz, T = 0.01).
    Characteristics: band-limited (|f| ≤ 50 Hz), even, real, non-causal, infinite duration,
zeros at t = ±0.01k, energy
    E = ∫|G|²df = 0.01²·100 = 0.01;  G(f)/T flat = the spectrum of a 'band-limited δ': as B →
∞, g → δ(t), G → 1
    — the Dirac delta's flat spectrum is the limiting case (justifies δ ⇔ 1).
    Samples zero except g(0) = A at Nyquist ⇒ g = A sinc(t/T)·…; same form scaled by A.
    Samples at 2B through an LPF (gain T, cutoff B): Σf(nT)δ(t − nT) → f(t) ⇒ output ∝ f(t)
(proportionality constant = filter gain × T... equals f when gain T).
    'Foretell the future': every future sample of a band-limited signal is determined by the
past (the sinc-sum       is a linear combination), but only in theory — infinite memory, exact
band limit and no noise are needed.
```

Lesson 12 — Practical Issues; Connection to the DFT

Lec 5 sl. 38–47

Time-limited vs Band-limited (sl. 38)

A signal cannot be both time-limited and strictly band-limited (uncertainty principle). Real signals are finite-duration, so their spectra have infinite tails; the tails always overlap in sampling ⇒ **some aliasing is unavoidable**. Practical measures: anti-aliasing filter, rate above Nyquist, realizable reconstruction filters (ZOH + analog LPF), quantization noise handled separately.

DTFT and DFT from the Sampled Signal (sl. 39–42)

- $X_p(j\omega) = \Sigma x[n]e^{-j\omega nT}$ repeats every ω_s; with $\omega T \to \omega$ this is the DTFT.
- The ideal LPF keeps one ω_s-wide period: $x(t) = (1/2\pi)\int_{-\omega_s/2}^{\omega_s/2} T\, X_p(j\omega)e^{j\omega t}d\omega$ ⇒ inverse DTFT at $t = nT$.
- The N-point DFT = N uniform samples of one ω_s-wide period: $X[k] = X_p(jk\omega_s/N)$.

Periodic Signals (sl. 43–47)

> $x(t) = \Sigma\, a_k\, e^{jk\omega_o t} \Rightarrow X(j\omega) = 2\pi\Sigma a_k \delta(\omega - k\omega_o)$; sample N times per period: $T_o = NT$, $\omega_s = N\omega_o$
> $X_p = (1/T)\Sigma X(j(\omega - r\omega_s)) = (2\pi/T)\Sigma\Sigma\, a_k\, \delta(\omega - k\omega_o - r\omega_s)$: every line is copied into every band
> Nyquist for periodic x: $a_k = 0$ for $|k| \geq N/2$ ($|k\omega_o| < \omega_s/2$) ⇒ one clean copy per band
> $c_k = a_k/T$, $X[k] = T_o c_k = N\, a_k$; inverse DFT $x[n] = (1/N)\Sigma_{k=0}^{N-1} X[k]e^{j2\pi kn/N}$

Practice — Lesson 12

(a) Actual — CSE 311 2013-14 Q2(b), 2020-21 Q3(a).

(a) What are the practical issues in sampling and reconstruction? Why can practical signals not be fully recovered? How are the problems handled/minimised? (10; 24)

✓ Solution

(a) Practical issues: (1) real signals are time-limited ⇒ not band-limited ⇒ aliasing from the spectral tails; (2) the ideal LPF/sinc interpolator is non-causal and infinite, so reconstruction is approximate (ZOH, linear interpolation, finite-length filters); (3) samplers are not ideal impulses (finite-width pulses, jitter); (4) quantization noise; (5) the signal must be known for all time for exact recovery. Handling: an anti-aliasing LPF before the sampler, sample above Nyquist (guard band), reconstruct with a ZOH followed by an analog low-pass (compensating the sinc droop), use practical band-limits (essential bandwidth), enough bits.

Quick Notes — Sampling

Revision points for Lessons 9–12.

L9 · Sampled spectrum

- $x_p = x\Sigma\delta(t - nT)$; $X_p = (1/T)\Sigma X(j(\omega - k\omega_s))$; $P = (2\pi/T)\Sigma\delta(\omega - k\omega_s)$.
- Duality: $X(jt) \leftrightarrow 2\pi x(-\omega)$; product $\leftrightarrow$ (1/2π) convolution.

L10 · Theorem, aliasing

- $\omega_s > 2\omega_M$; $T < \pi/\omega_M$. Overlap = aliasing: $f \rightarrow |f - kf_s|$; irreversible.
- Cure: anti-alias LPF, higher rate. Product xy: bandwidths add. Natural sampling: sinc-weighted copies.
- DT: $M \leq \pi/W$ ($W = 2\pi/9 \Rightarrow M = 4$).

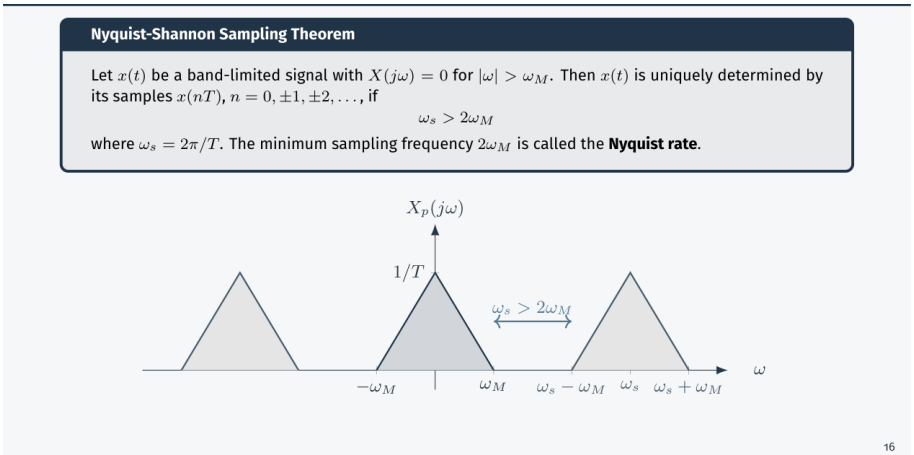
L11 · Reconstruction

- Ideal: $H = T$ on $|\omega| < \omega_s/2$, $h = \text{sinc}(t/T)$, $x = \Sigma x(nT)\text{sinc}((t - nT)/T)$.
- ZOH: $h_0 = \text{rect}[0,T]$, $H_0 = Te^{-j\omega T/2}\text{sinc}(\omega T/2\pi)$. Linear: h_1 = triangle, $H_1 = T\,\text{sinc}^2(\omega T/2\pi)$.
- Real-time DAC → ZOH.

L12 · Practice

- Time-limited ⇒ not band-limited ⇒ some aliasing always; $X[k] = N a_k$.

Key Diagrams to Reproduce



The sampling theorem picture (Lec 5, sl. 20)